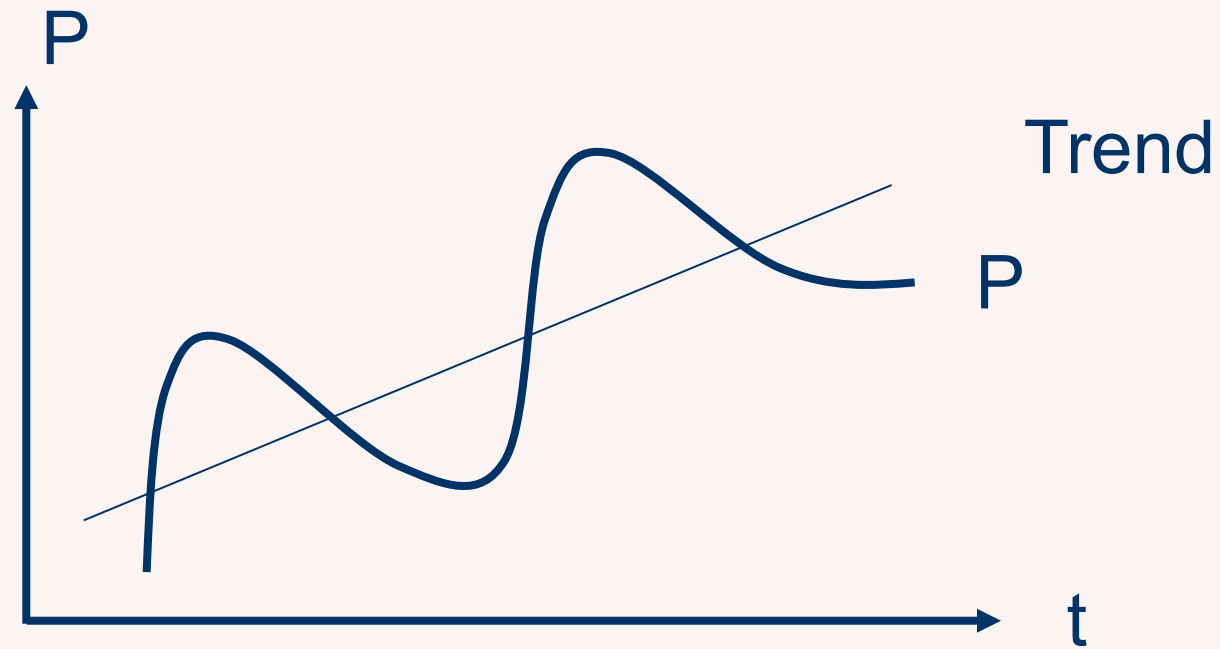


Block II—Topic 5:
**Real Estate Cycles
Driven by Exogenous Demand Shocks**



Last lecture

- Looked at phenomenon of real estate cycles
 - ▶ Are 'local' in nature
 - ▶ Cyclicity varies a lot across markets
- Market demand curve is downward sloping and hence long-run supply price elasticity matters for price dynamics
- Supply curve is kinked suggesting asymmetric response to positive and negative demand shocks

Open questions (Discuss)

This lecture: Real Estate Cycles Driven by Exogenous Demand Shocks

- The role of long-term supply constraints: The case of England
 - a) Different types of long-run supply constraints
 - b) Endogeneity concerns & identification strategy (IV-approach)
 - c) Empirical results for England
 - d) Counterfactual analysis for England
- The role of credit conditions

This lecture: Real Estate Cycles Driven by Exogenous Demand Shocks

- The role of long-term supply constraints: The case of England
 - a) Different types of long-run supply constraints
 - b) Endogeneity concerns & identification strategy (IV-approach)
 - c) Empirical results for England
 - d) Counterfactual analysis for England
- The role of credit conditions

One open question...

- ⇒ What types of constraints make supply inelastic?
- ⇒ Consider the case of England...

(Based on Hilber & Vermeulen, 2016)

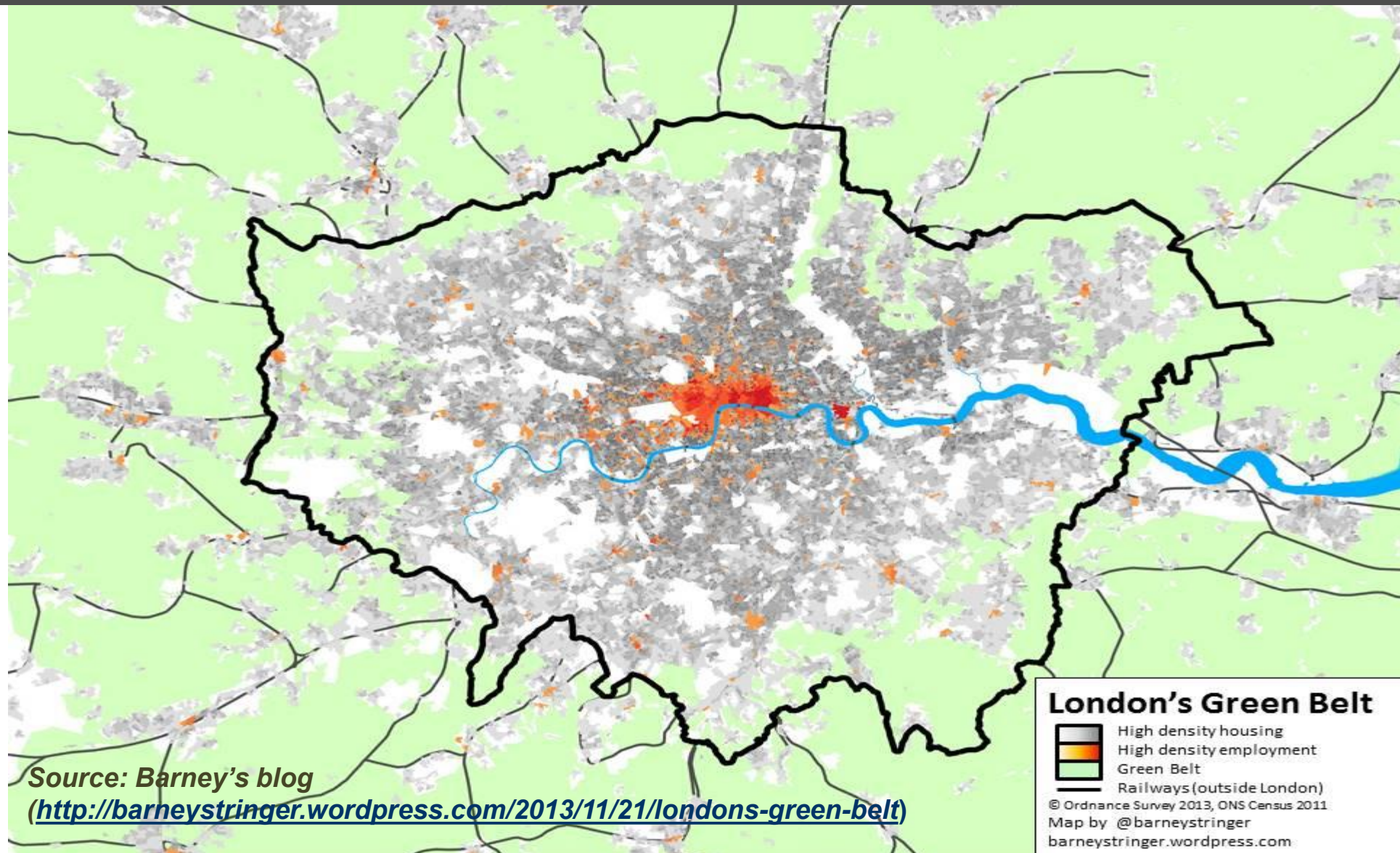
(An application of an IV-approach)

Candidate #1:

Regulatory supply constraints

- English planning system widely viewed as inflexible
 - ▶ Since 1947: virtually no fiscal incentives at local level to permit development
 - ▶ ‘Development control system’ (catering to NIMBYs) particularly near green belts
 - ▶ ‘Horizontal’ constraints: Green belts surrounding major cities
 - ▶ ‘Vertical’ constraints: height restrictions & protected vistas
 - ▶ Other regulations: preservation policies (conservation areas, listed buildings) & codes

An illustration of London's restrictiveness: 1. London's green belt



2. London's height restrictions, preservation policies & protected vistas



Historic preservation

Conservation areas



Listed buildings

Example of early “international style”
architecture (built 1935)
(Architect: Berthold Lubetkin)



Candidate #1:

Regulatory supply constraints (cont.)

- Barker-review (2004, 2006) suggested that regulatory constraints *may be* important *causal* driver of high house prices and volatility

⇒ To be tested...

Candidate #2:

Physical supply constraints

- Could also be physical supply constraints
 - ▶ a) **Limited local availability of open developable space**
(very high opportunity costs)
 - ▶ b) **Steep slopes** (difficult + costly to build)
- ⇒ To be tested...

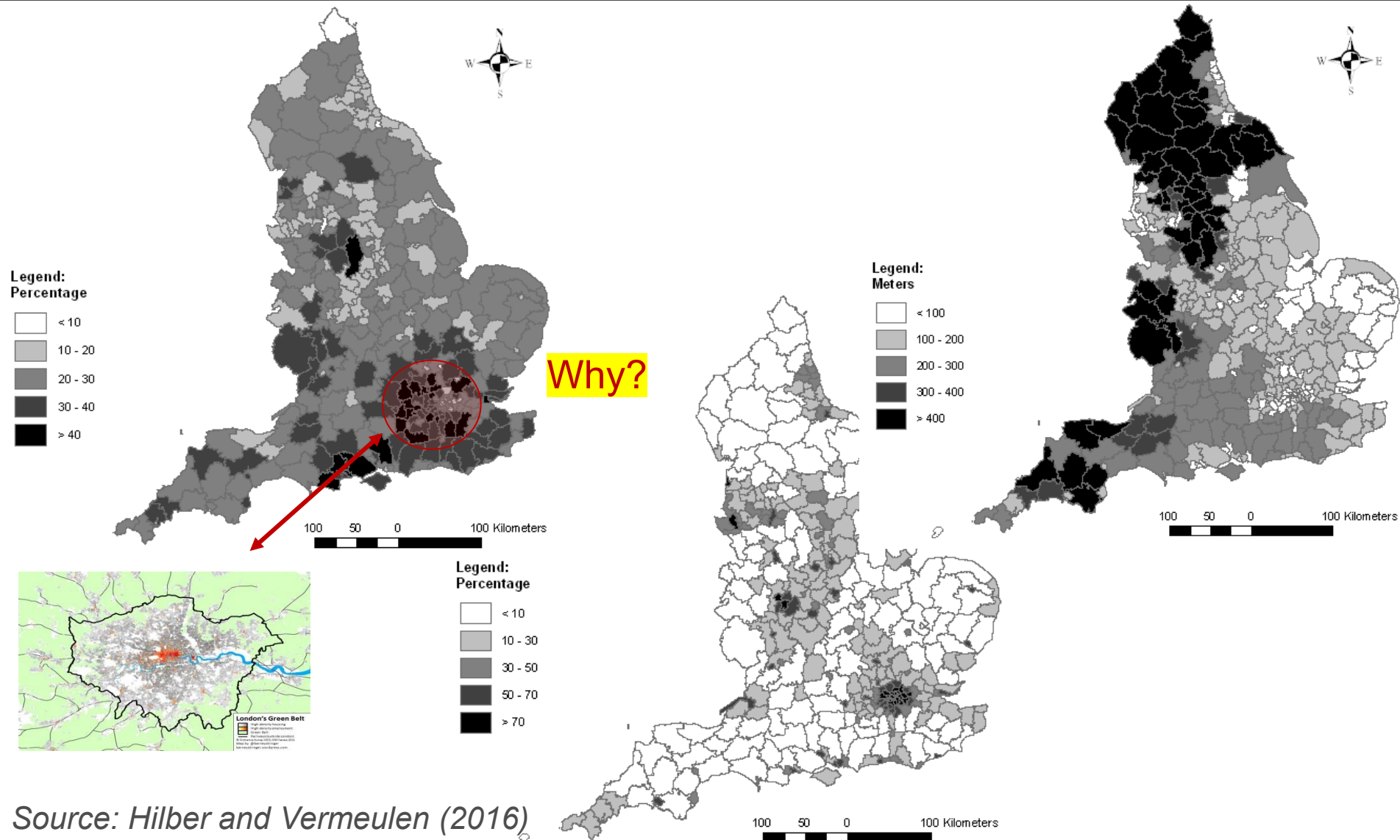
How to test in practice?

- Proxy for regulatory constraints
 - ▶ Use direct measure of how restrictive **Local Planning Authorities (LPAs)** are: **Refusal rate for major residential projects** (1979-2008)
- Proxies for physical constraints
 - ▶ Use land cover satellite data to calculate **share developable land that is developed** (in 1990)
 - ▶ Use raster grid data to derive measure of slope related constraints: **range in elevation** (or alternatively: standard deviation of slope)
- All three measures are 'standardized' to ease interpretation of coefficients ($x^{stand.} = (x - \bar{x})/\sigma$)

Average refusal rate
(major residential
projects) 1979-2008

Share developable
land developed,
1990

Elevation range



How to test rigorously?

Empirical strategy

- Estimating equation:

$$\begin{aligned} \log(\text{house price}_{jt}) &= \beta_0 + \beta_1 \log(\text{earnings}_{jt}) + \\ &\quad \beta_2 \log(\text{earnings}_{jt}) \times \text{refusal rate}_j + \\ &\quad \beta_3 \log(\text{earnings}_{jt}) \times \% \text{developed}_j + \\ &\quad \beta_4 \log(\text{earnings}_{jt}) \times \text{elevation}_j + \text{year} - FE \\ &\quad + LPA - FE + \varepsilon_{j,t} \end{aligned}$$

$j = 1, \dots, 353$
 $t = 1974, \dots, 2008$

$\beta_2, \beta_3, \beta_4 > 0$

Empirical strategy (cont.)

- Three potential endogeneity concerns:
 - ▶ **Refusal rate:** Refusal rate may be endogenous to demand conditions & developers may not apply if likely rejected
 - ▶ **Share developed:** Contemporaneous D & S factors (incl. regulation) may affect share developed
 - ▶ **Earnings:** Local earnings can be influenced by house prices (via sorting) and therefore may reflect housing supply as well as housing demand + likely correlated with construction sector wages
- ⇒ Problem: Estimates of ordinary regressions are likely biased
- ⇒ Luckily, instrumental variables (IV) approach (2SLS) allows us to address this problem & identify unbiased causal effects...

Basic idea of IV (2SLS) approach (repetition from Topic 3)

- Find ‘instrumental variables’
 - ▶ Ideally strongly correlated with endogenous RHVs (here: **refusal rate, share developed & earnings**), conditional on other covariates
 - ▶ But do not directly impact the LHV (here **house prices**) (uncorrelated with error term)
- Use exogenous variation from instruments to predict endogenous RHVs (refusal rate, share developed & earnings) in **1st stage**
- Then use predicted RHVs in **2nd stage** to identify the causal and unbiased effect of these RHVs on the LHV (house prices)

IVs to identify causal effect of refusal rate

Two instruments:

1. Change in delay rate (pre-/post-policy reform in 2002) [IV#1]
 - ▶ Labour government introduced delay rate targets in 2002, but no refusal rate targets!
 - ▶ Restrictive LPAs had strong incentive to substitute delays with refusals
 - ▶ Most restrictive LPAs will be ones with greatest decrease of delay rate post-reform
 - ▶ **Identifying assumption:** Conditional on location FEs, change in delay rate affects impact of earnings on house prices only through planning restrictiveness

IVs for refusal rate (cont.)

2. Local vote share [IV#2]

- ▶ Middle income Labour voters have traditionally cared more about housing affordability and less about protecting house values (fewer own homes!)
- ▶ **Identifying assumption:** Conditional on location FEs, local vote share affects impact of earnings on house prices only through planning restrictiveness

IV to identify causal effect of land scarcity

- Instrumental variable: Historical density from 1911 [IV#3]
 - ▶ Instrument pre-dates ‘birth’ of modern British planning system (TCPA of 1947) by several decades
 - ▶ **Identifying assumption:**
 - Density almost 100 years ago will be indicative of early forms of agglomeration & local amenities, so should be strongly correlated with share developed land today
 - But, controlling for LPA FEs, historic density should not directly explain *changes* in contemporaneous HPs

IV to identify causal effect of earnings

- Instrumental variable: ‘labor demand shock’ measure (Bartik 1991) [IV#4]
 - ▶ Use local industry composition in 1971 and national employment growth in the industries to predict local employment growth (shift-share approach)
 - ▶ Local industry composition in 1971 pre-dates our regression sample
 - ▶ Ideally: would instrument earnings (with LDS) but leads to weak identification
 - ▶ Replace earnings with plausibly exogenous demand shock measure (can no longer interpret coefficients as price-earnings elasticity)

Empirical strategies

- Strategy 1: Use TSLS and instruments #1 to #3 to identify causal effects of supply constraints measures & ignore concern that local earnings might be endogenous
- Strategy 2: Replicate this specification but replace earnings with instrument #4
 - ▶ Can no longer interpret coefficient as price earnings elasticity
 - ▶ But yields plausibly unbiased estimates

Data

- House price index
 - ▶ **Land Registry** (1995 - 2008), **CML** (1974 - 1995)
 - ▶ Index adjusts for mix of housing types
 - Real weekly earnings of FT working men
 - ▶ **ASHE / NES**
 - + Regulatory data (DCLG), satellite data (various sources), historic data (Census)
- ⇒ All geographically matched to 2001 LPA boundaries (353)

Results: naïve OLS

LHV: Log(real house price index)	
	Coefficient
Log(earnings)	0.32***
Log(earnings) x <i>average refusal rate</i>	0.067***
Log(earnings) x <i>share developed</i>	0.094**
Log(earnings) x <i>elevation range</i>	-0.00047
Year-FEs	Yes
LPA-FEs & constant	Yes

House price-earnings elasticity of LPA with average constraints

Refusal rate +1 std. dev. (+8.7%) → price-earn. elasticity increases by +0.067 (~+21%)

Endogenous!

Results of IV: 1st stage (validity of instruments)

- All three ‘instruments’ (#1 - #3) have the predicted sign and are highly statistically significant (at 1%-level)
- Test-statistics suggest that instruments may be valid and strongly identify the causal effects of regulatory and scarcity related supply constraints

Results of IV: 2nd stage

LHV: Log(real house price index)	
Log(earnings)	0.089
Log(earnings) x <i>average refusal rate</i>	0.29***
Log(earnings) x <i>share developed</i>	0.30***
Log(earnings) x <i>elevation range</i>	0.095**
Year-FEs	Yes
LPA-FEs & constant	Yes
Kleibergen Paap F-stat	11.8

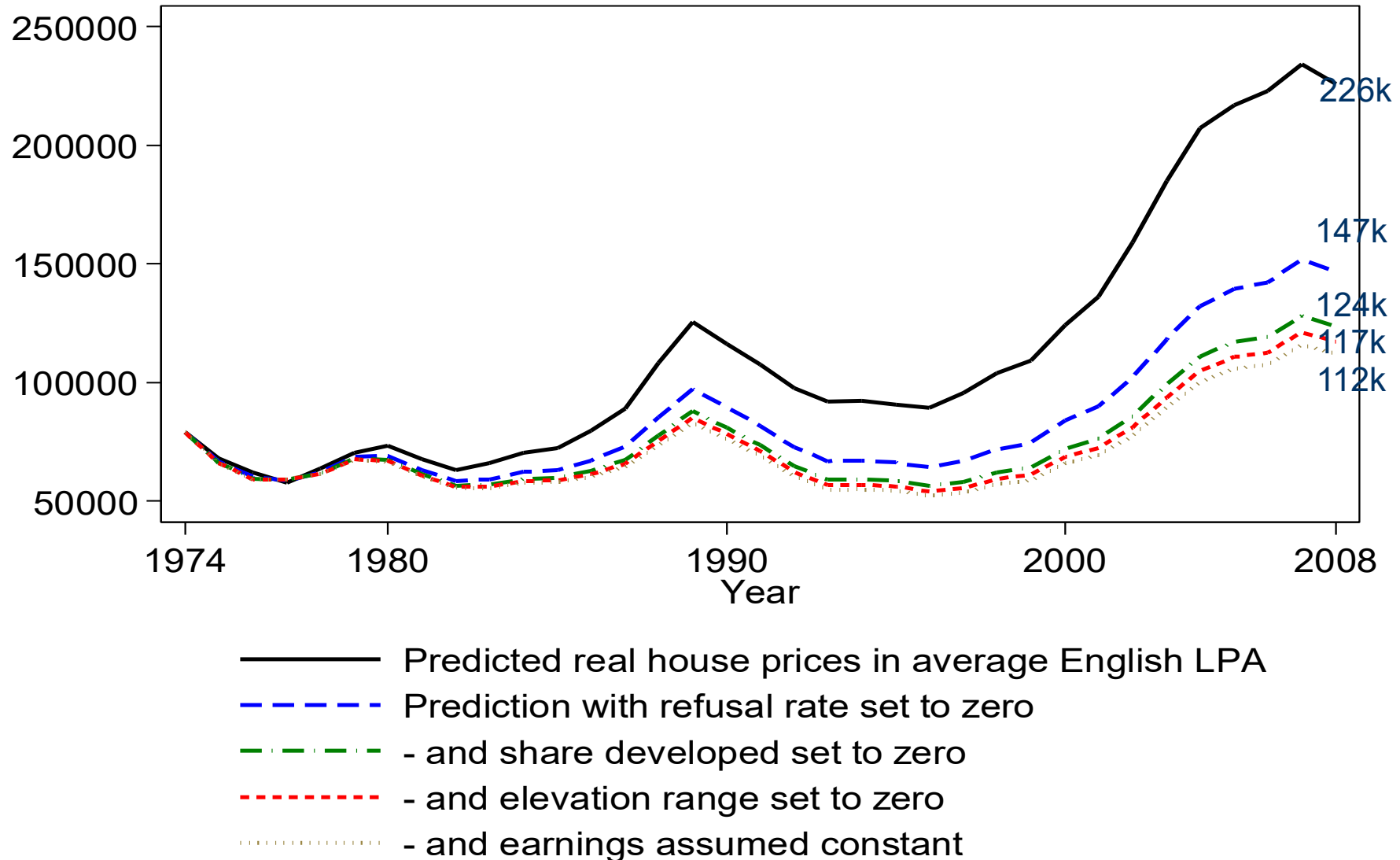
Quantitative effects

(based on IV with instruments #1-3)

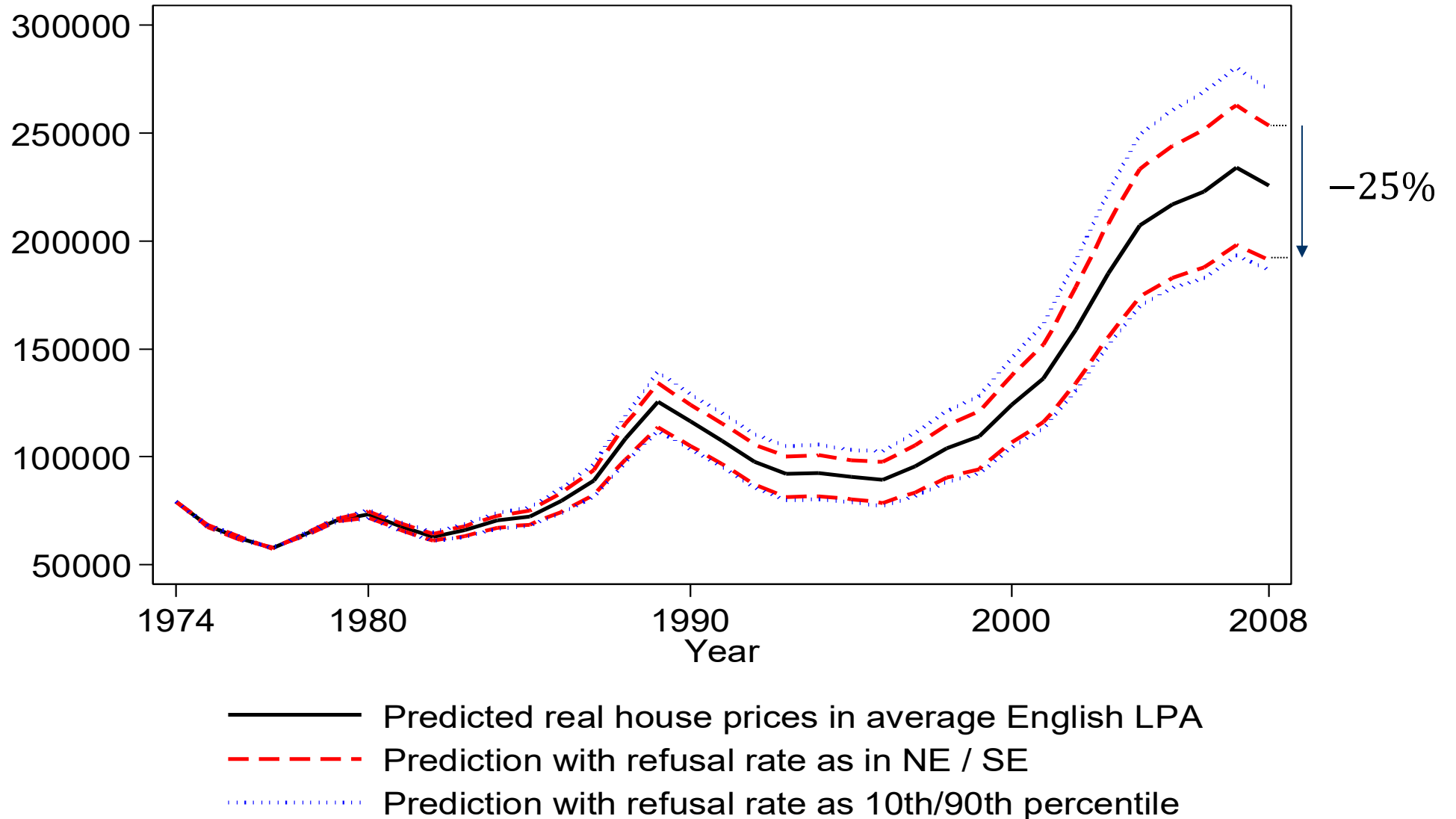
- If **planning system** were **relaxed** in *av. LPA*:
 - ▶ House prices in *av. LPA*: **-35%**
- *and* **developable land** were **abundant**:
 - ▶ House prices in *av. LPA*: **-45%**
- *and* LPA were **completely flat**:
 - ▶ House prices in *av. LPA*: **-48%**

Note: These are likely lower bound estimates for several reasons (see paper for details)

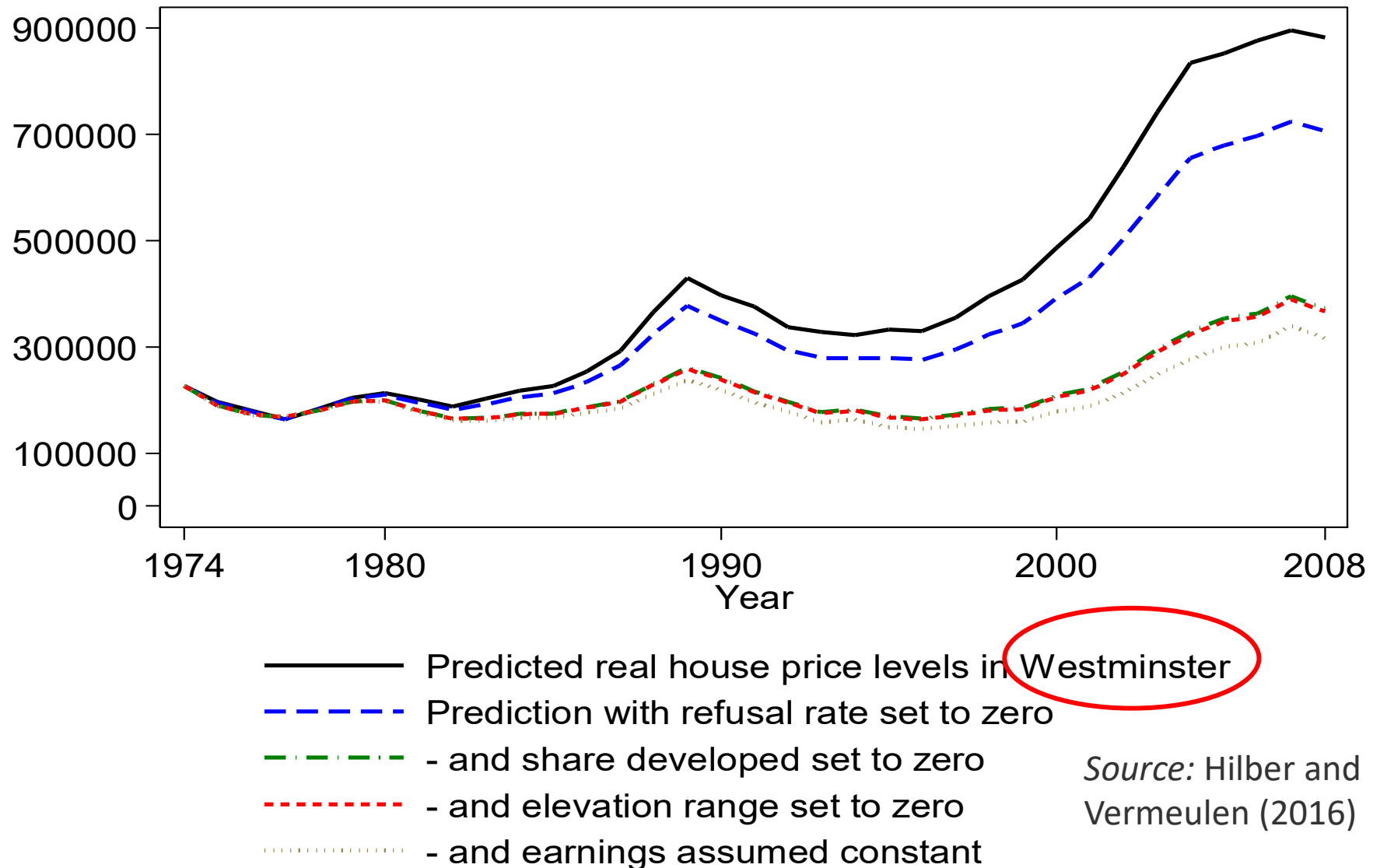
What would house prices in average English LPA be if...



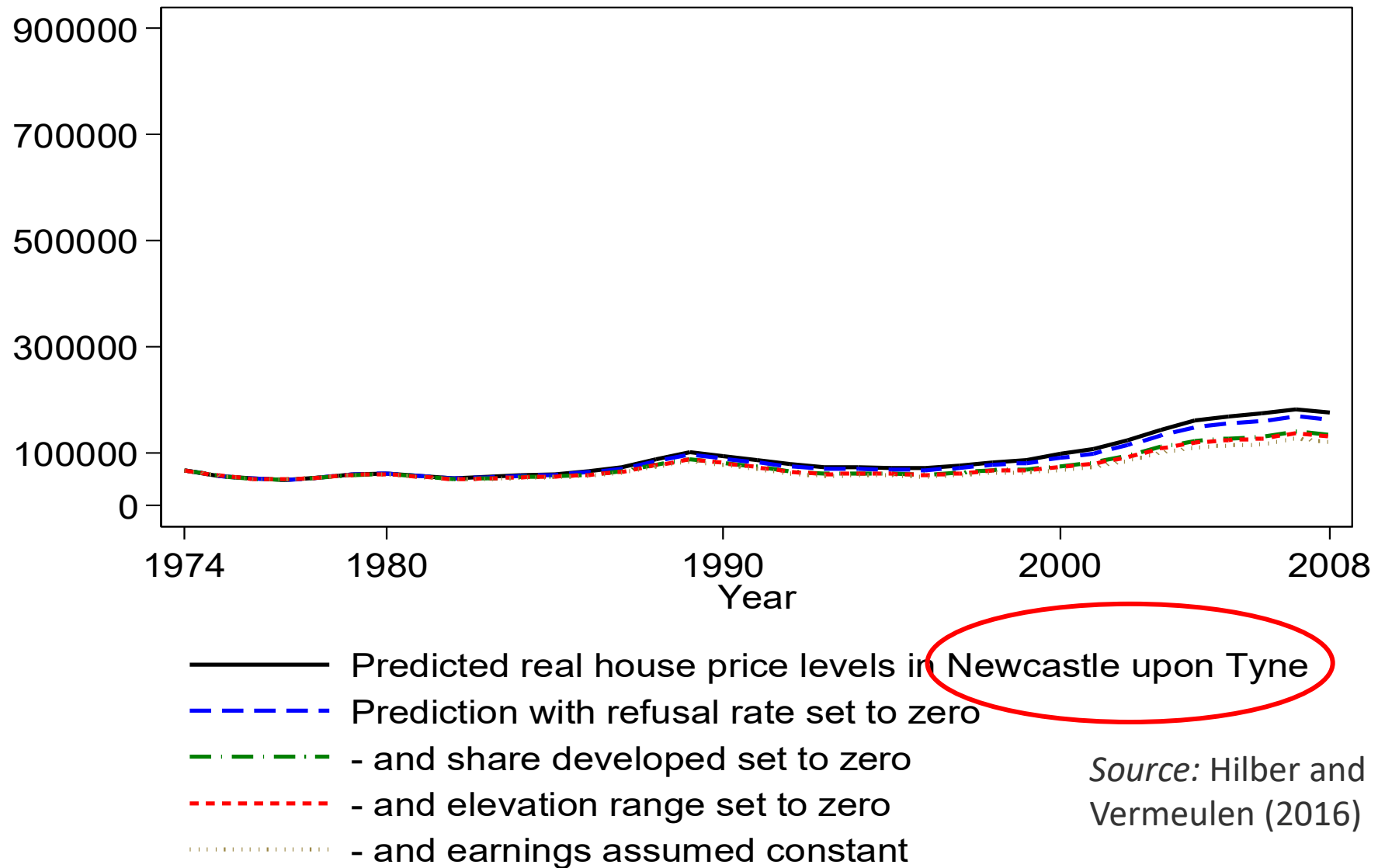
North East vs. South East & 90th vs. 10th percentile



The importance of supply constraints varies across markets: Westminster (London)



House price dynamics in Newcastle



Use labour demand shock instead of earnings (IV 2nd stage)

LHV: Log(real house price index)		
	LPA	TTWA
<i>Log(labour demand shock)</i>	0.31**	0.24**
<i>Log(LDS) x average refusal rate</i>	0.66***	0.59***
<i>Log(LDS) x share developed</i>	0.92***	0.39***
<i>Log(LDS) x elevation range</i>	0.33**	0.12
Year-FEs	Yes	Yes
LPA-FEs & constant	Yes	Yes
Kleibergen Paap F-stat	5.2	65.7

Other results & robustness checks

- Differential impact of supply constraints significantly larger during boom than during bust
 - ▶ Consistent with kinked supply curve
- Impact of local land scarcity confined to highly developed locations (GLA)
- Main results hold for alternative definitions of 'local housing markets' (TTWAs, urban TTWAs, FUR, Pre-1996 counties)

Robustness checks (Cont.)

- Results not sensitive to using alternative measures of share developed (excl. semi-developable land; flood risk areas)
- Results not sensitive to using alternative proxies for elevation/ruggedness
- Results hold for alternative IV-strategies and alternative measure for regulatory restrictiveness (shadow price)

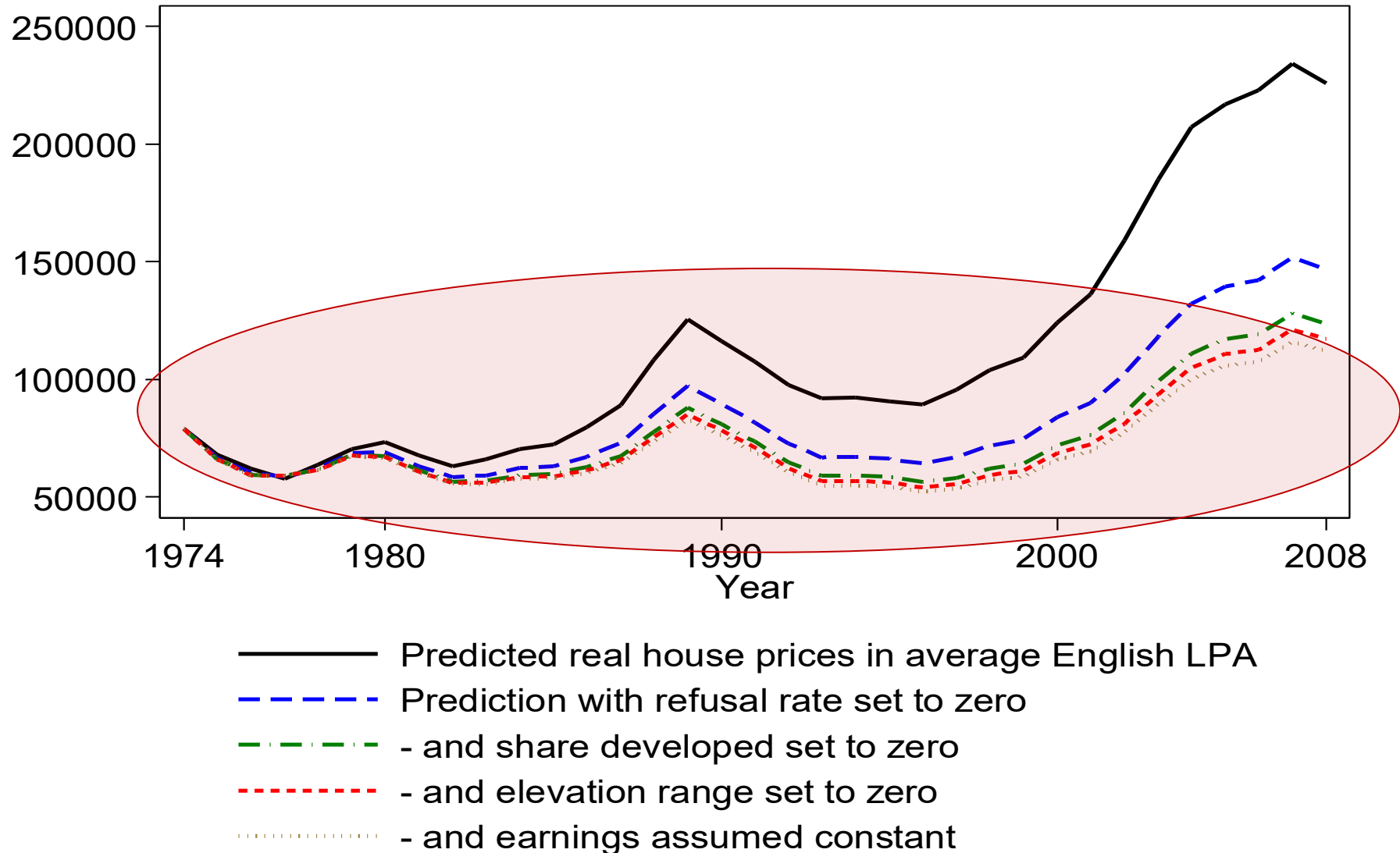
Preliminary conclusions

1. **Regulatory constraints in conjunction with strong demand in desirable areas are main causal driver of severe housing affordability crisis & volatility in parts of UK**
 2. **Physical constraints** matter too but impact is very **non-linear**
 3. All local supply constraints and earnings fluctuations jointly still **cannot explain all cyclical**
- ⇒ Suggests role for macro-economic factors (interest rates, credit conditions)...

This lecture: Real Estate Cycles Driven by Exogenous Demand Shocks

- The role of long-term supply constraints: The case of England
 - a) Different types of long-run supply constraints
 - b) Endogeneity concerns & identification strategy (IV-approach)
 - c) Empirical results for England
 - d) Counterfactual analysis for England
- The role of credit conditions

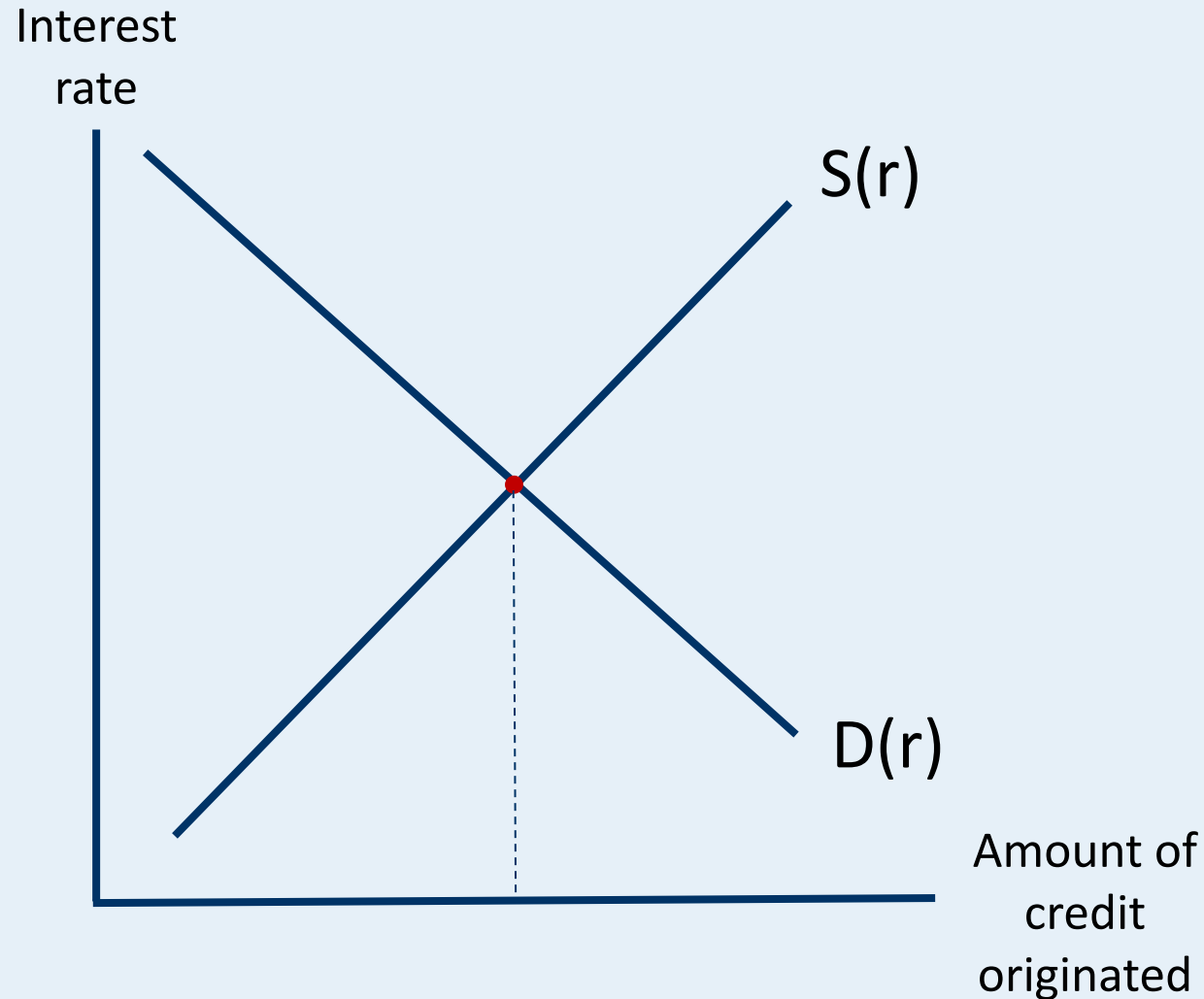
How can we explain remaining variation in house price cyclicalty?



Role of credit conditions?

- In Hilber & Vermeulen (2016)
 - ▶ **Year-FEs** capture impact of all time-varying factors at macro-level (including credit supply and/or demand shocks)
 - ▶ But year-FEs are a 'black box' – **can't tell what factors** are driving fluctuations at macro-level
 - ▶ Also, sample period **stops in 2008** and misses most recent years with credit crunch and expansion
- ⇒ Look next at studies that focus on role of credit conditions...

How do credit conditions affect house prices? (Discuss)



Credit conditions and house prices

- Numerous studies explore effects of equilibrium credit on house prices
 - ▶ *Examples:* Mian & Sufi (2009), Adelino *et al.* (2012), DiMaggio & Kermani (2014)
- Find positive effects on house prices during build-up of Great Financial Crisis
- Problem: Changes in securitization (expansion of subprime) or conforming loan limits are not exogenous due to
 - ▶ **Reverse causation** (HPs serve as collateral) or
 - ▶ **OV-bias** (credit may respond endogenously to current and expected economic conditions)

Favara and Imbs (2015)

- Idea: Exploit branching deregulation (lifting of branching restrictions) in US b/w 1994-2005 as exogenous instrument for credit supply
 - ▶ Explore **causal effect** of exogenous expansion in mortgage credit on
 - Origination of mortgage loans and
 - House prices

Main findings of Favara and Imbs (2015)

1. Deregulation explains one-half to two-thirds of observed increase in mortgage loans originated by commercial banks & **one-third to one-half of increase in house prices**
2. Placebo tests rule out demand-based explanations (i.e., lower collateral requirements that shift demand for credit)
 - ▶ Credit supply shock causes lower interest rates (whereas shift of demand for credit would put upward pressure on interest rates) [*see Justiniano et al. 2019*]
3. Credit supply shocks increase HPs less in areas with elastic housing supply (supply constraints still important!)

Were credit supply shocks similarly important in UK? (Discuss)

Preparations for **Seminar 5** (Topics 4/5/6, Week 9)

- **Topic:** Property Market Dynamics: Theory and Empirical Tests
- **Seminar format:** Group presentations of answer to past exam question & subsequent discussion
- **Key readings** for seminar:
 - a. Compulsory readings for Topics 5/6 (pick two theories/papers; e.g., Case, Shiller & Thompson, 2012 & Hilber and Vermeulen, 2016)
 - b. Capozza *et al.* (2004)
 - c. DeFusco *et al.* (2018), Nathanson & Zwick (2018)
- **Details:** See separate handout (on Moodle)

Preparations for **Seminar 6** (Topic 7, Week 10)

- **Topic:** Should governments subsidize homeownership?
- **Seminar format:** Key paper presentations (see handout on Moodle)
- **Papers to be discussed:** DiPasquale and Glaeser (1999) and Hilber and Turner (2014)
- **Tasks for all students:** Read key papers and prepare questions for presenters